Project Design Phase
Proposed Solution Template

Date	03 April 2026
Team ID	SWTID-2026-8629
Project Name	Smart Library
Maximum Marks	2 Marks

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1. Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation that shows how data flows through a system. It helps in understanding how input data is transformed into output through processes.

Key Components of DFD:

1. External Entity – Source or destination of data (e.g., Student, Librarian)
2. Process – Activity that transforms data (e.g., Issue Book, Return Book)
3. Data Store – Storage of data (e.g., Library Database)
4. Data Flow – Movement of data between entities

Level 0 DFD (Context Diagram)

Description:
This diagram represents the overall system as a single process and shows interaction with external entities.

Entities:

- Student
- Librarian

Process:

- Smart Library System

Data Flows:

- Student searches and requests books
- Librarian manages books and approvals
- System provides availability status and reports

Level 1 DFD

Processes:

1. Add Book Details
2. Update Book Information
3. Issue Book
4. Return Book
5. View Book Records
6. Generate Reports

Data Store:

- Library Database

Flow Explanation:

- Librarian adds/updates book data → Stored in database
- Student requests book → Librarian issues book → Database updated
- Book return → Availability updated
- System retrieves data → Displays reports and status

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2. User Stories

User stories describe system requirements from the user's perspective.

Format:
"As a [user], I want [function], so that [benefit]."

User Stories for Smart Library System

1. **Book Search**
    - As a student, I want to search for books so that I can find required materials easily.
2. **View Book Details**
    - As a student, I want to view book availability so that I know if the book is available.
3. **Add Book**
    - As a librarian, I want to add new books so that the library database is updated.
4. **Update Book Information**
    - As a librarian, I want to update book details so that records remain accurate.
5. **Issue Book**
    - As a librarian, I want to issue books to students so that borrowing is tracked properly.
6. **Return Book**
    - As a librarian, I want to record returned books so that availability is updated.

7. **Generate Reports**
    - As a librarian, I want to generate reports so that I can analyze library usage.
8. **Delete Book Record**
    - As a librarian, I want to remove outdated book records so that the database stays clean.

Conclusion:

The Data Flow Diagram provides a clear visualization of how data moves within the Smart Library system, while user stories help in understanding user requirements. Together, they ensure a well-structured and user-centered system design.